Mehmood Sheikh *AI Engineer*

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PROFESSIONAL SUMMARY

AI Engineer with year of hands-on experience in Python, Machine Learning, and AI systems. I've worked on real-world projects involving AI agents using the Eliza AI Framework, developing voice-based AI system, implementing RAG pipelines for custom knowledge, and fine-tuning voice models (TTS) that solve business problems. I'm passionate about learning new technologies, building practical solutions, and contributing to teams focused on innovation and impact.

PROFESSIONAL EXPERIENCE

A.I. Specialist 07/2025 – Present | Onsite - Full Time
Intellexal Solutions Private Limited

- **Designed the Lumina core system connecting a standard phone network to an advanced conversational AI**, enabling the company to deploy intelligent, **voice-based agents over phone calls**.
- **Built an N8N-style Agentic AI System in Lumina** that connects **chatbots to 20+ business tools (Google Sheets, CRMs, APIs) allows AI to perform actions** like saving data to Google Sheets, checking order status, booking appointments etc.
- **Developed the AI processor to accurately listen to callers**, understand what they mean, generate smart responses, and speak back with a natural, human-like voice.
- **To enable sophisticated, human-like automated conversations**, I engineered an end-to-end voice **AI pipeline (VAD → STT → LLM → TTS)** with **dynamic persona templates**, empowering the agent to conduct nuanced, role-specific dialogues.
- **Enabled the AI to provide highly accurate, client-specific answers** by building a custom knowledge base for each client using a **Weaviate vector database** and an **information retrieval (RAG) pipeline**.
- **To create bilingual (Urdu/English) voice for the AI**, Currently **fine-tuning a Text-to-Speech (TTS) model** on **Common Voice dataset**, eliminating the robotic sound for a more natural user experience.
- **To ensure system reliability and provide key business insights**, I implemented a **PostgreSQL data persistence layer** to log all **call records, full transcripts, and cost metrics,** creating a complete and auditable trail for every interaction.

AI/ML Developer 11/2024 – 06/2025 | Remote - Contract
FameGuild

- **Faced with slow marketing decision-making,** I designed an AI agent to generate marketing strategies, forecast trends, and answer key business questions — **reducing planning time by 40% and improving campaign accuracy.**
- **To boost brand visibility across platforms,** I built an automation tool that created and scheduled social media posts with personalized captions & dynamic images, smart hashtags, and ideal timing — **leading to a 60% increase in engagement and saving 30+ hours per week.**
- **To support NFT model training,** I developed a pipeline that automatically collected and cleaned NFT images and metadata — **accelerating dataset preparation by 5× and ensuring seamless input for text-to-image AI models.**
- **With growing data volume,** I integrated modern tools to improve processing and feature extraction — **boosting data handling capacity by 10× and enabling real-time model updates.**

Junior ML & Python Developer 06/2024 – 08/2024 | Remote - Contract
Freelance

- **Built multiple ML models** (Logistic Regression, KNN, Decision Tree, Random Forest, ANN) — **achieved 99.44% accuracy** using Random Forest after applying **SMOTE, RFE, and ANOVA**.
- **Preprocessed large datasets**, handling missing values and encoding variables to ensure clean, usable input for model training.
- **Automated geospatial workflows** using **Python, GeoPandas, and Pandas**, enabling efficient processing of spatial data.
- **Created spatio-temporal visualizations** including **3D maps and MP4 animations** with QGIS and Python to showcase trends in spatial data.
- **Delivered final outputs** as **Jupyter notebooks, animations, and a research paper**, meeting all project and publication standards.

EDUCATION

XSEED Incubation Program 2.0 10/2025 – Present | Karachi, Pakistan
INSEAD
XSEED is an entrepreneurship hub powered by researchers from INSEAD Business School.

Bachelor of Science in Software Engineering 02/2021 – 02/2025 | Karachi, Pakistan
NED University of Engineering & Technology
CAMPUS: Usman Institute of Technology (Affiliated with NED)

CGPA: 3.157

CERTIFICATES

- Fundamentals of AI Agents Using RAG and LangChain | Coursera
- Mathematics for Machine Learning: Linear Algebra | Coursera
- Full-Stack Deep Learning with Python | LinkedIn Learning
- PyTorch Essential Training: Deep Learning | LinkedIn Learning
- Recurrent Neural Networks | LinkedIn Learning
- Python Data Structures | Coursera
- Programming for Everybody (Getting Started with Python) | Coursera

SKILLS

Technical
Python, Fast API, SQL, Postgres.

Python Libraries
Pytorch, Tensorflow, Keras, Sklearn, Matplotlib, Numpy, Pandas, Seaborn, Huggingface, Transformers.

PROJECTS

Rahbar 10/2025 – Present
The Web Spoken Digital Guider

- **To make the web accessible to millions in Pakistan facing literacy barriers,** I designed and built "Rahbar," a conversational AI that transforms the visual web into a fully interactive audio experience, enabling users to browse, search, and understand content using only their voice.
- **Engineered an intelligent page analysis agent** that identifies all interactive elements (links, forms, buttons) and provides step-by-step vocal guidance, effectively translating complex visual layouts into simple, actionable audio commands (e.g., "Say 'click contact us' to proceed").
- **To expand Rahbar into a full productivity partner,** I integrated Google Suite APIs (Gmail, Calendar) and local file system operations, empowering users to manage emails, check schedules, and perform file CRUD tasks purely through conversation.

AI-Powered Job Finder 04/2025 – 04/2025

- **To automate job discovery**, I **developed a scalable job search API** using **async scraping** from platforms like **LinkedIn, Indeed, and Rozee.pk**, enabling real-time job aggregation across multiple sources.
- **Integrated Google Gemini via LangChain** to deliver **AI-based job relevance scores (0–100)** — with **fallback logic** ensuring consistent results even during partial service failures.
- **Designed a multi-layered scraping engine** using **API calls, HTML parsing, and Google Search fallback**, with **user-agent rotation** to bypass blocks and ensure stability.
- **Built a modular and testable system** with **unit + integration tests**, CI/CD compatibility, and **live deployment via Railway** using environment-based configurations.

FYP | HIREXCEL 02/2024 – 10/2024

- **Faced with inefficiencies in traditional hiring**, I developed an **AI-powered assessment platform** that combined **cognitive, personality, and technical evaluations** — delivering real-time analytics and **precise candidate profiling**.
- **To personalize job recommendations**, I built **AI models** that analyzed candidate traits and skills — providing recruiters with a **synopsis of personality, intrapersonal abilities, and cognitive strengths**, aligned with role requirements.
- **Automated job posting workflows** to enable **real-time personality and skills matching**, helping recruiters make faster, data-driven hiring decisions.
- **Integrated predictive algorithms** to **rank candidates based on job fit and growth potential**, improving recruitment accuracy and **reducing manual screening time**.

LANGUAGES

English ●●●●○ **Urdu** ●●●●●

IELTS ACADEMIC

- CEFR Level : C1
- Overall Band Score : 7.0
- TRF No. : 25PK502450SHEM363A